

SIMRAN KAUR

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PROJECTS

StudySphere AI – AI-Powered Full-Stack Learning Platform | Link

Jul – Aug 2026

- Engineered a full-stack MERN application (React, Vite, Node.js, Express.js, MongoDB Atlas) with JWT authentication and protected routes, exposing **44 REST API endpoints across 10 modules** covering auth, documents, subjects, flashcards, quizzes, AI chat, analytics, and study planning.
- Architected a **Retrieval-Augmented Generation (RAG) pipeline** using **LangChain**, the Google **Gemini API**, and MongoDB Atlas **Vector Search** — chunking and embedding uploaded PDF documents to power context-aware AI chat, summarization, flashcard, and quiz generation.
- Designed document upload and management (**Multer**) and subject-organization features backed by Mongoose schemas, plus a study planner and analytics dashboard for tracking learning activity.
- Structured the backend into modular controllers, services, **middleware**, and routes for maintainability; deployed both frontend and backend to production on Vercel.

Tech: MERN Stack, JavaScript, Vite, LangChain, Google Gemini API, RAG, MongoDB Atlas Vector Search, JWT, Multer, Axios

Pehchaan – Voter ID Verification System | Link

Mar – May 2026

- Developed the **backend services** for a full-stack voter verification platform using **Node.js, Express.js, MongoDB, and JWT**, implementing REST APIs, authentication/authorization, and data handling.
- Built and integrated the **Python-based ML service** using **FastAPI, Pandas, and Scikit-learn** to process voter records and support automated identity verification achieving **92% classification accuracy** on the evaluation dataset.
- Integrated the backend with the ML service to enable **end-to-end voter verification**, connecting frontend requests, backend APIs, database operations, and ML-based validation.
- Collaborated with a **5-person team** on backend architecture, API integration, authentication, and **ML-service** workflows using Git/GitHub.

Tech: React.js, Node.js, Express.js, MongoDB, Python, FastAPI, Pandas, Scikit-learn, JWT, REST APIs, Axios, Machine Learning

Event Management System | Link

May 2026

- Architected a **JavaFX-based** event management application with **role-based access control** for admin and user workflows.
- Integrated **BCrypt** authentication with **MySQL/JDBC** to secure user credentials and manage persistent relational data.
- Engineered **CRUD** functionality following **MVC architecture and OOP principles**, separating UI, business logic, and database operations.
- Configured **Maven**-based dependency management and maintained version-controlled development using **Git**, following modular design and maintainable coding practices.

Tech: Java, JavaFX, MySQL, JDBC, Maven, BCrypt, MVC, OOP, Git

CERTIFICATES

- Java Programming — neocolab Jun 2026
- Software Engineering — Saylor Academy Sep 2025

SKILLS

Programming Languages: Java, Python, JavaScript

Frontend: React.js, HTML5, CSS3, React Router

Backend: Node.js, Express.js, FastAPI

AI/ML: RAG, LangChain, Google Gemini API, Pandas, Scikit-learn

Databases: MongoDB, MongoDB Atlas, MySQL

Developer Tools: Git, GitHub, VS Code, IntelliJ, Maven, Postman

Concepts: RESTful API Design, JWT Authentication, Data Structures & Algorithms, OOP, MVC Architecture

Soft Skills: Adaptability, Team Collaboration, Problem Solving, Quick Learner

ACHIEVEMENTS

- Solved **200+ LeetCode problems** across **Data Structures and Algorithms**.
- Dean's List** — **Top 10%**, Lovely Professional University.

EDUCATION

Lovely Professional University

Master of Computer Applications — CGPA: 7.96

2025 – Present

Phagwara, Punjab

Lovely Professional University

Bachelor of Computer Applications — CGPA: 8.05

2022 – 2025

Phagwara, Punjab